Classification and Clustering Project

Bank Marketing: Subscription Prediction and Customer Segmentation

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Batch: PGDA0226 | Machine Learning Phase 6 - Level 2

Dataset: UCI Bank Marketing - bank-additional-full.csv
41,188 rows | 21 columns | Target: y

Executive Summary

This project analyses the UCI Bank Marketing dataset to support two connected tasks: predicting whether a client will subscribe to a term deposit and discovering natural customer/campaign segments. The full dataset contains 41,188 rows and 21 columns. The target is imbalanced but workable: 4,640 records (11.27%) are 'yes' and 36,548 (88.73%) are 'no'.

Three classification algorithms were trained on the same stratified 80/20 split using a preprocessing pipeline with scaling and one-hot encoding. The call-duration field was excluded because it is only known after the call and would create information leakage for a realistic pre-call prediction. Random Forest achieved the best positive-class F1 score among the three compared models and was selected for detailed evaluation.

K-Means clustering was then applied without the target using scaled numeric features. Values of k from 2 to 6 were compared using inertia and silhouette score; k=3 had the strongest silhouette score and produced three interpretable segments. Integration analysis showed that the classifier behaves very differently across these segments, which is more actionable than using overall accuracy alone.

1. Dataset and Business Problem

1.1 Dataset source and coverage

Dataset: Bank Marketing (with social/economic context), created by Sergio Moro, Paulo Cortez and Paulo Rita. The full file covers direct marketing contacts from May 2008 to November 2010 and includes five social/economic indicators. Source: UCI Machine Learning Repository, Bank Marketing dataset. The supplied documentation cites Moro, Cortez and Rita (2014), 'A Data-Driven Approach to Predict the Success of Bank Telemarketing', Decision Support Systems.

1.2 Business question

Can a bank use customer, campaign-history and economic-context information to identify clients who are more likely to subscribe to a term deposit, while also identifying natural segments that reveal where the prediction model performs differently?

1.3 Target and end user

The target column is y, where 'yes' means the client subscribed to a term deposit and 'no' means the client did not. The intended end user is a marketing or campaign manager responsible for prioritising outreach, controlling repeated contacts and allocating campaign resources.

1.4 Suitability against project requirements

Requirement	Dataset evidence
Real published dataset	UCI Bank Marketing; documented public research dataset
At least 1,000 rows	41,188 rows

At least 8 usable columns	21 columns in the source file
Categorical target	Binary target y: yes/no
No tiny target class	Minority yes class = 11.27%
Enough numeric features	Age, campaign, pdays, previous and five macroeconomic indicators support clustering

2. Exploratory Data Analysis

2.1 Structure, missing values and duplicates

The original file has 41,188 rows and 21 columns. There are no standard NaN values. The source documentation explains that some missing categorical values are encoded as the literal label 'unknown'. These were retained as a separate level rather than guessed. The dataset contained 12 exact duplicate rows, which were removed before modelling.

Column	unknown count
job	330
marital	80
education	1731
default	8597
housing	990
loan	990

2.2 Target balance

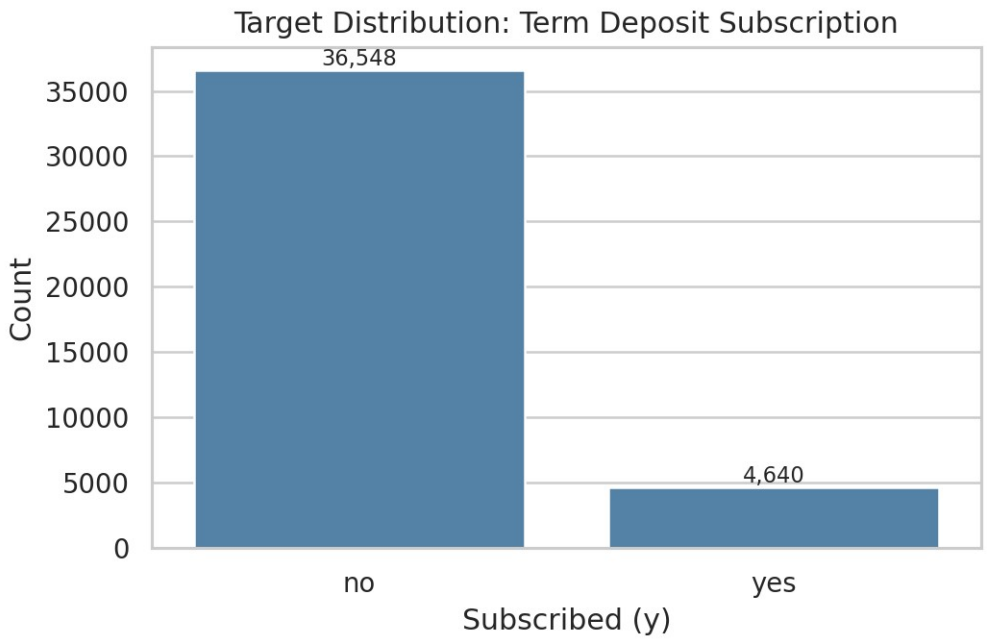


Figure 1. Target class distribution.

The majority class is 'no' (88.73%) and the positive 'yes' class is 11.27%. A majority-only classifier would therefore look strong on accuracy despite having no value for finding subscribers. This is why precision, recall and F1 are reported for the positive class.

2.3 Customer patterns

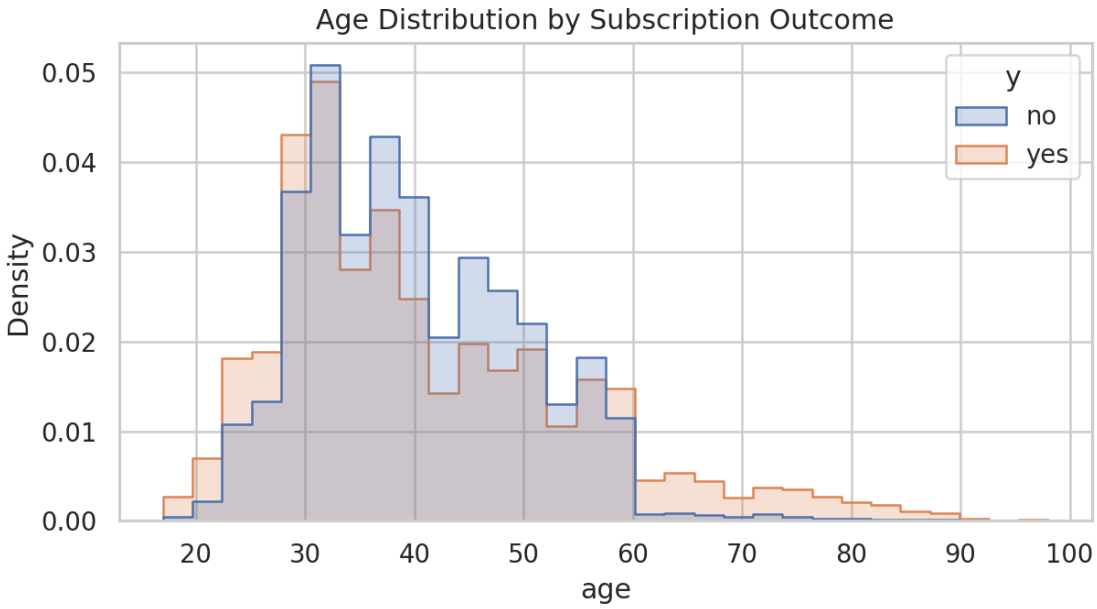


Figure 2. Age distribution by subscription outcome.

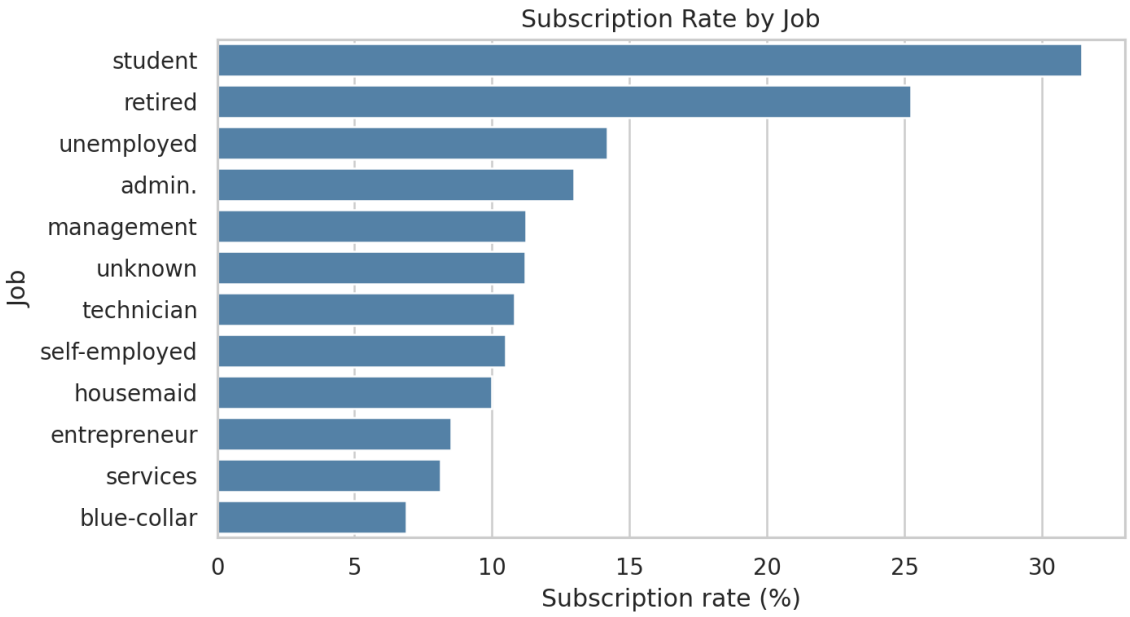


Figure 3. Subscription rate by job category.

Response rates differ across job categories and age distributions are not identical between outcomes. This supports keeping both numeric and categorical customer attributes in the classifier.

2.4 Numeric relationships

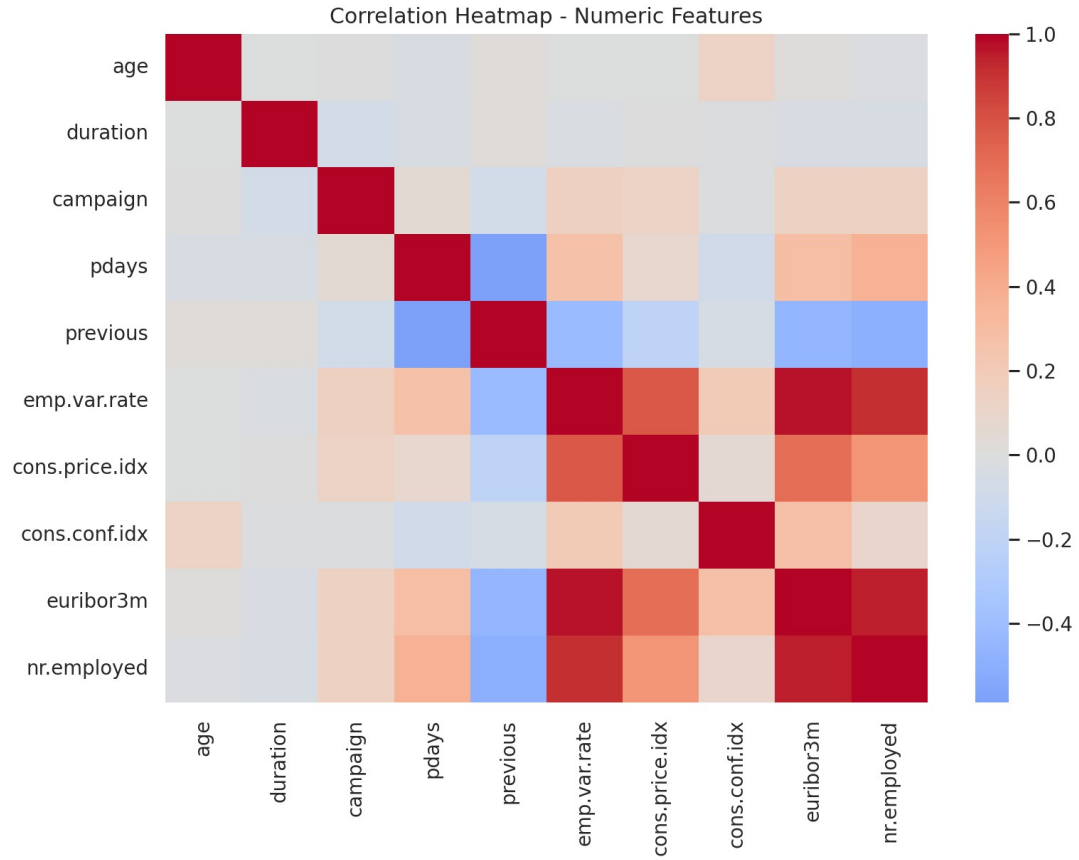


Figure 4. Correlation heatmap for numeric variables.

Several economic variables are strongly correlated because they represent the same macroeconomic environment. They are retained because the project compares multiple algorithms on the same split and Random Forest is not dependent on a linear-independence assumption.

3. Data Preparation

- Exact duplicate rows were removed before modelling.
- The source-coded 'unknown' category was retained for categorical variables.
- Call duration was excluded from the realistic predictive feature set because it is known only after the call has occurred.
- Categorical features were converted using one-hot encoding.
- Numeric features were scaled using StandardScaler inside a preprocessing pipeline.
- The data was split 80/20 using stratification and random_state=42 before any model was trained.
- The same split and preprocessing approach were used for all classifiers to make comparison fair.

The preprocessing pipeline also contains median and most-frequent imputers so that the workflow remains robust if standard missing values appear in later versions of the file, even though the supplied file has no NaN values.

4. Classification Model

4.1 Baseline

The majority-class baseline accuracy on the test set is 88.73%. This means a model can exceed 88% accuracy without identifying any subscribers at all. The main comparison therefore focuses on positive-class F1, supported by precision and recall.

4.2 Model comparison

Algorithm	Accuracy	Precision	Recall	F1
Random Forest	0.893	0.548	0.288	0.377
Decision Tree	0.841	0.316	0.353	0.334
Logistic Regression	0.898	0.650	0.210	0.318

Random Forest was selected because it achieved the best F1 score (0.377) among the compared models. Logistic Regression produced slightly higher accuracy but weaker recall/F1 for the positive class, showing why accuracy alone is not the right selection rule here.

4.3 Confusion matrix and metric interpretation

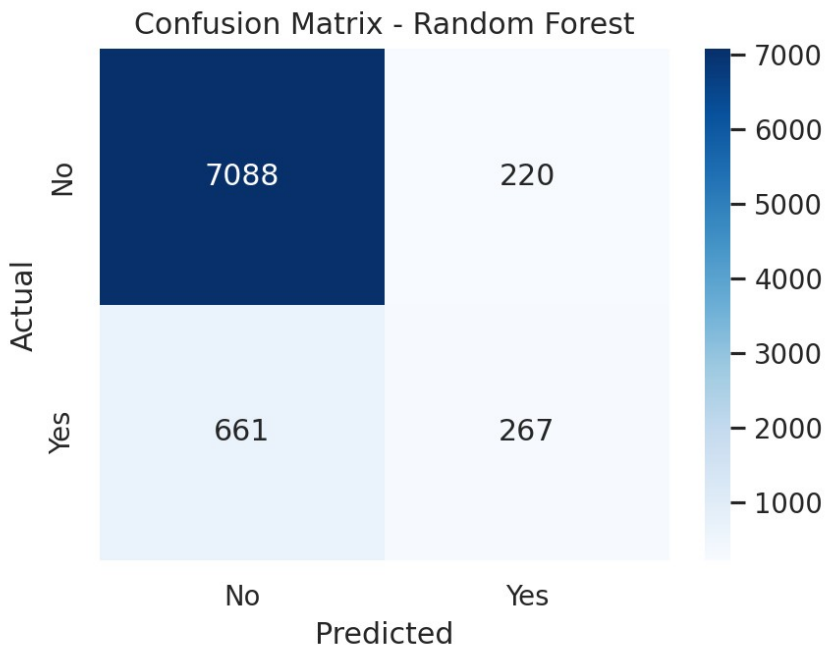


Figure 5. Confusion matrix for the selected Random Forest model.

The test confusion matrix contains 7,088 true negatives, 220 false positives, 661 false negatives and 267 true positives. The selected model has accuracy 0.893, precision 0.548, recall 0.288 and F1 0.377.

For the campaign use case, recall for the 'yes' class is especially important because a false negative is a potentially receptive customer who is not identified by the model. The relatively low recall shows that the default model should be treated as a decision-support tool rather than an automatic replacement for campaign judgement.

4.4 Feature importance

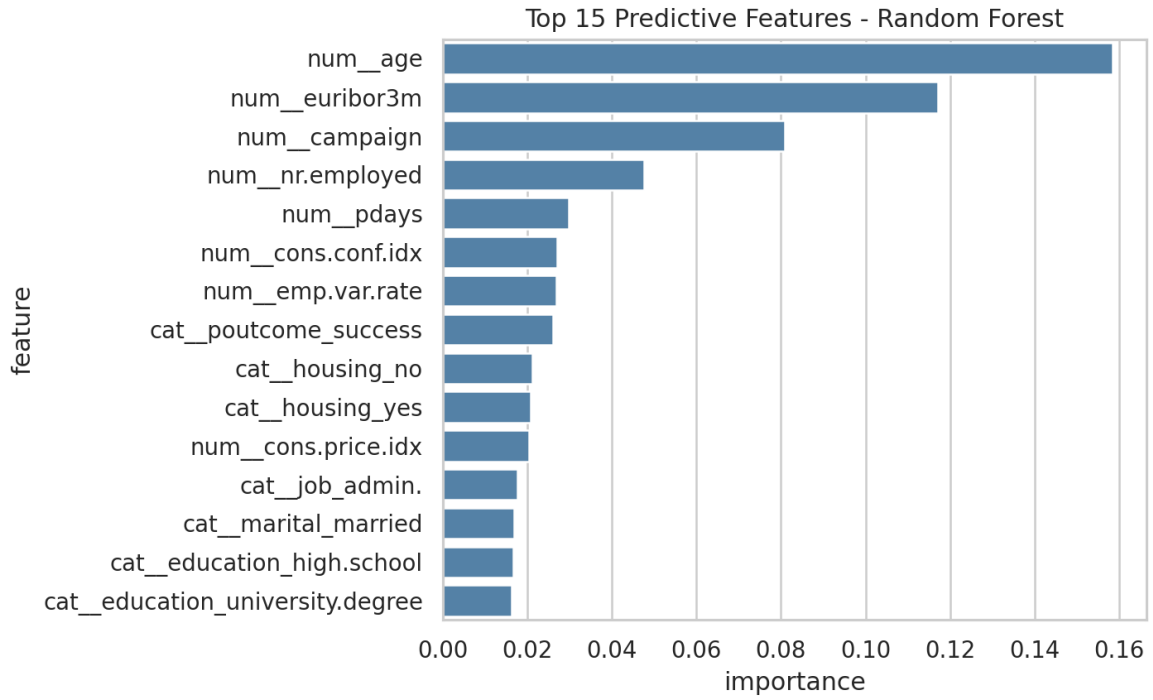


Figure 6. Top Random Forest feature importances.

Rank	Feature	Importance
1	num__age	0.1584
2	num__euribor3m	0.1172
3	num__campaign	0.0808
4	num__nr.employed	0.0476
5	num__pdays	0.0299
6	num__cons.conf.idx	0.0270
7	num__emp.var.rate	0.0268
8	cat__poutcome_success	0.0260
9	cat__housing_no	0.0211
10	cat__housing_yes	0.0207

Age, Euribor, campaign activity, employment levels, previous-contact timing and prior campaign outcome appear among the strongest contributors. These are predictive associations, not causal effects.

5. Clustering

5.1 Features and scaling

K-Means was run on nine scaled numeric features: age, campaign, pdays, previous, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m and nr.employed. The target y was not supplied to K-Means, and duration was excluded for consistency with the realistic pre-call use case.

5.2 Choosing k

k	Inertia	Silhouette
2	225,686	0.474
3	175,048	0.488
4	153,158	0.457
5	132,383	0.470
6	117,575	0.315

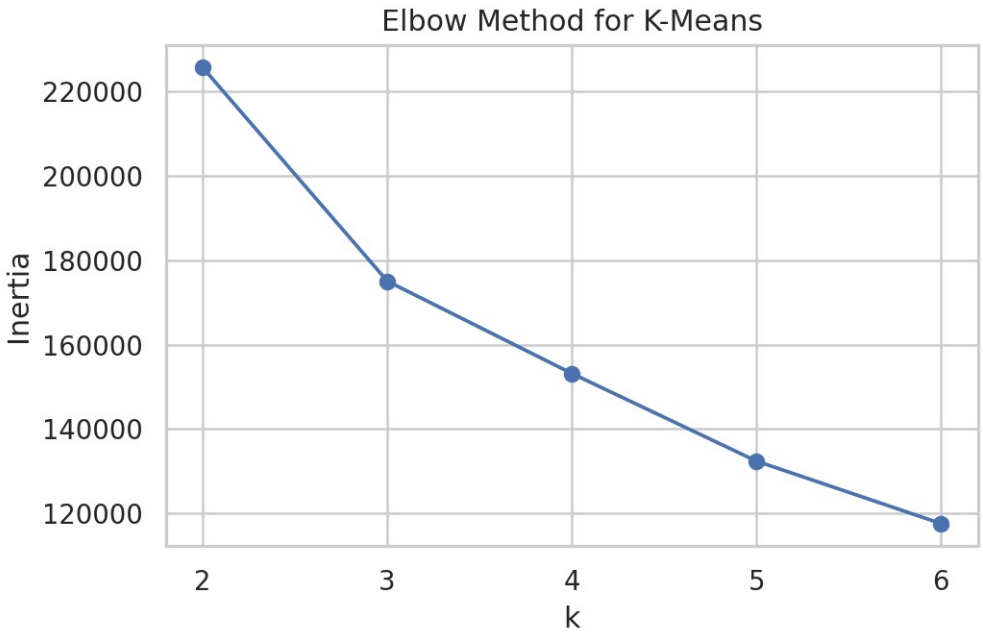


Figure 7. Elbow curve for k=2 to 6.

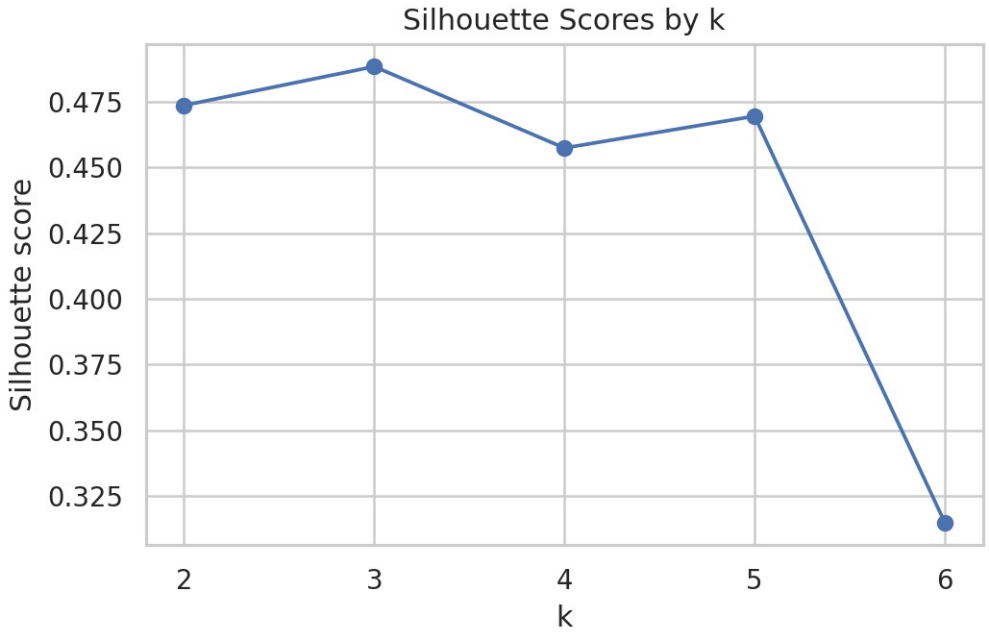


Figure 8. Silhouette scores for k=2 to 6.

k=3 was selected because it produced the highest silhouette score (0.488) and generated a small set of segments that can be clearly interpreted by a campaign manager.

5.3 Cluster visualisation and profiles

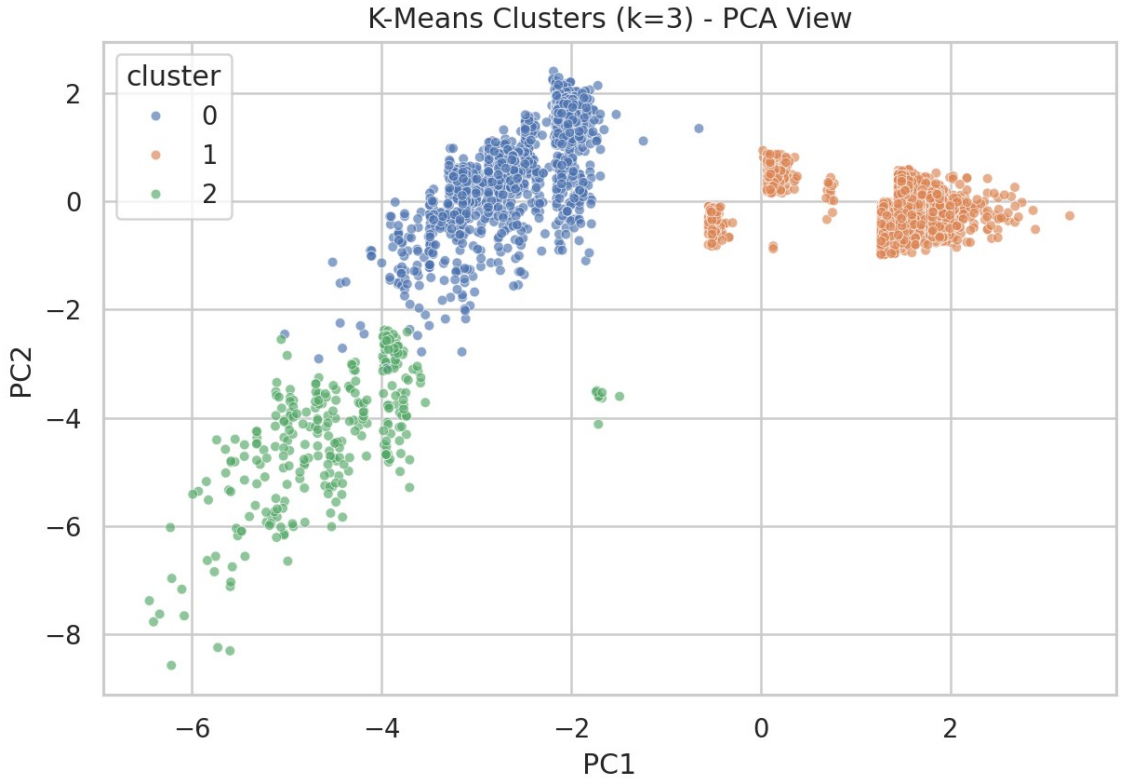


Figure 9. PCA visualisation of the three K-Means clusters.

Cluster	Segment	Size	Share	Yes rate	Campaign	Previous	Pdays
0	Mainstream Low-Engagement Segment	12001	29.15%	19.44%	2.13	0.32	999.00
1	High-Contact Campaign Segment	27637	67.12%	4.83%	2.80	0.03	999.00
2	Previously Engaged - Higher Response	1538	3.74%	63.13%	1.83	1.69	20.86

Cluster 1 is the largest and has the lowest subscription rate, despite a comparatively high number of current-campaign contacts. Cluster 2 is small but stands out for previous engagement, low pdays and a very high subscription rate. Cluster 0 sits between the two and represents a broader low-engagement segment in a lower-rate environment.

6. Integration of Classification and Clustering

6.1 Cluster relationship with the actual target

The clusters were created without using y, but their observed subscription rates differ sharply. This means the unsupervised structure is related to the business outcome. The strongest natural segment is the previously engaged group, while the large high-contact segment has the weakest conversion rate.

6.2 Model performance within each cluster

Cluster	Segment	Test n	Accuracy	Recall yes	F1 yes
0	Mainstream Low-Engagement Segment	2374	0.789	0.220	0.302
1	High-Contact Campaign Segment	5548	0.951	0.036	0.062
2	Previously Engaged - Higher Response	314	0.659	0.798	0.737

The model's overall accuracy hides strong segment-level differences. Accuracy is highest in the large low-response cluster because most records are non-subscribers. Positive-class recall and F1 are strongest in the previously engaged high-response cluster. This makes segment-level monitoring essential.

6.3 Combined business conclusion

A campaign manager should prioritise the previously engaged segment because it combines high observed propensity with much stronger positive-class detection. The high-contact, low-response segment should be reviewed for contact efficiency, because additional contact attempts do not correspond to high conversion. The classifier can support prioritisation, but its low overall positive recall means it should not be used as the sole screening mechanism. Performance should be tracked by cluster rather than only by overall accuracy.

7. Limitations and Responsible Interpretation

- The target is imbalanced, so default classifiers favour the majority class.
- Hyperparameter tuning was deliberately not used; the project focuses on correct workflow and interpretation.
- Feature importance does not establish causation.
- The data covers 2008-2010, so relationships may differ in a modern banking environment.
- K-Means assumes distance-based, roughly spherical grouping and is sensitive to the selected feature space.
- The PCA chart is only a two-dimensional visual summary and does not contain all clustering information.

8. Reproducibility

The notebook uses random_state=42 for the split, tree-based models, K-Means and PCA. The supplied notebook was executed top-to-bottom with outputs saved. The dataset is stored in the same folder, so the notebook uses a relative path and can be rerun without editing the file location.

9. References

UCI Machine Learning Repository. Bank Marketing dataset. Source page: https://archive.ics.uci.edu/dataset/222/bank+marketing

Moro, S., Cortez, P., & Rita, P. (2014). A Data-Driven Approach to Predict the Success of Bank Telemarketing. Decision Support Systems. DOI: 10.1016/j.dss.2014.03.001.

Banco de Portugal statistical indicators referenced in the dataset documentation.